

Computer Science & Information Technology

Discrete Mathematics

DPP: 1

Combinatorics

Q1 Let B_n represent the number of bit strings of length n that contains a substring "00". Write the recurrence relation for B_n .

- (A) $B_n = B_{n-1} + B_{n-2} + 2^{n-2}$
 (B) $B_n = B_{n-1} + B_{n-2} - 2^{n-2}$
 (C) $B_n = B_{n-1} - B_{n-2} + 2^{n-2}$
 (D) $B_n = B_{n-1} - B_{n-2} - 2^{n-2}$

Q2 A population of bacteria has initial count 200. After one hour, the population reached 220. The population grows in such a way that the number of additional bacteria per hour, doubles each hour.

Write a recurrence relation ' b_n ' to describe the number of bacteria after n hours.

We know $b_0 = 200$ and $b_1 = 220$

- (A) $b_n = 2b_{n-1} + 3b_{n-2}$
 (B) $b_n = 2b_{n-1} - 3b_{n-2}$
 (C) $b_n = 3b_{n-1} + 2b_{n-2}$
 (D) $b_n = 3b_{n-1} - 2b_{n-2}$

Q3 Find the solution of recurrence relation

$$a_n = 2a_{n-1} + 3 \quad n \geq 1$$

with initial terms $a_0 = 2$

- (A) $5 \cdot 2^n + 3$ (B) $5 \cdot 2^n - 3$
 (C) $3 \cdot 2^n - 5$ (D) $3 \cdot 2^n + 5$

Q4 Solve the recurrence relation $a_n = a_{n-1} + 8a_{n-2} - 12a_{n-3}$, subjected to the initial conditions $a_0 = 0$, $a_1 = 1$, $a_2 = 3$.

- (A) $a_n = \left(\frac{1}{25} + \frac{2n}{5}\right)2^n - \left(\frac{2}{25}\right)(-3)^n$
 (B) $a_n = \left(\frac{1}{25} + \frac{2n}{5}\right)2^n - \left(\frac{1}{25}\right)(3)^n$
 (C) $a_n = \left(\frac{1}{25} - \frac{2n}{5}\right)2^n - \left(\frac{1}{25}\right)(-3)^n$
 (D) $a_n = \left(\frac{1}{25} + \frac{2n}{5}\right)2^n - \left(\frac{1}{25}\right)(-3)^n$

Q5 The correct option is/are with respect to the recurrence relation $a_n = 11a_{n-1} - 30a_{n-2}$, subjected to the initial conditions $a_0 = 1$, $a_1 = 5$.

(A) The characteristic roots are 0 and 1.

(B) $a_n = 5^n$

(C) The characteristic roots are 5 and 6.

(D) $a_n = 1 \cdot 5^n$

Q6 Solve the recurrence relation $a_n + 7a_{n-1} - 18a_{n-2} = 2^n + 1$, subjected to the initial conditions $a_0 = 1$, $a_1 = 2$.

- (A) $a_n = \frac{4n}{11}2^{n-1} - \frac{1}{10}$
 (B) $a_n = \frac{4n}{11}2^{n-1} + \frac{1}{10}$
 (C) $a_n = \frac{4n}{11}2^{n-2} - \frac{1}{10}$
 (D) $a_n = \frac{3n}{11}2^{n-1} - \frac{1}{10}$

Q7 Solve the recurrence relation $a_n = 6a_{n-1} - 9a_{n-2} + 3^n$, subjected to the initial conditions $a_0 = 1$, $a_1 = 2$.

- (A) $a_n = 3^{n-2} [9 + 3n + {}^nC_2]$
 (B) $a_n = 3^{n-2} [9 - 3n - {}^nC_2]$
 (C) $a_n = 3^{n-2} [9 - 3n + {}^nC_2]$
 (D) $a_n = 3^{n-2} [9 + 3n - {}^nC_2]$

Q8 Solve the recurrence relation $a_n - 2a_{n-1} + a_{n-2} = n$, subjected to the initial conditions $a_0 = 1$, $a_1 = 2$.

- (A) $a_n = 1 + \frac{n}{3} + \left(\frac{n}{6} + \frac{1}{2}\right)n^2$
 (B) $a_n = 1 - \frac{n}{3} + \left(\frac{n}{6} + \frac{1}{2}\right)n^2$
 (C) $a_n = 1 + \frac{n}{3} - \left(\frac{n}{6} + \frac{1}{2}\right)n^2$
 (D) $a_n = 1 + \frac{n}{3} + \left(\frac{n}{6} - \frac{1}{2}\right)n^2$

Q9 Solve the recurrence relation $a_n - 3a_{n-1} - 10a_{n-2} = (2n+1) \cdot 5^n$, subjected to the initial conditions $a_0 = 1$, $a_1 = 2$.

- (A) $a_n = \left(\frac{156}{133}\right)5^n + \frac{23}{133}(-2)^n + 5^n \left(\frac{n}{38} + \frac{5}{38}\right)n$
 (B) $a_n = \left(\frac{156}{133}\right)5^n - \frac{23}{133}(-2)^n - 5^n \left(\frac{n}{38} + \frac{5}{38}\right)n$
 (C) $a_n = \left(\frac{156}{133}\right)5^n - \frac{23}{133}(-2)^n + 5^n \left(\frac{n}{38} + \frac{5}{38}\right)n$


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$$(D) a_n = \left(\frac{156}{133}\right)5^n - \frac{23}{133}(2)^n + 5^n \left(\frac{n}{38} + \frac{5}{38}\right)n$$

Q10 Solve the recurrence relation $a_n = 11a_{n-1} - 24a_{n-2} + 3^n$, subjected to the initial conditions $a_0 = 1, a_1 =$

2.

$$(A) a_n = \left(\frac{77}{73}\right)3^n - \frac{4}{73}(8)^n - \frac{n}{5}(3)^{n-1}$$

$$(B) a_n = \left(\frac{77}{73}\right)3^n + \frac{4}{73}(8)^n - \frac{n}{5}(3)^{n-1}$$

$$(C) a_n = \left(\frac{77}{73}\right)3^n - \frac{4}{73}(8)^n + \frac{n}{5}(3)^{n-1}$$

$$(D) a_n = \left(\frac{77}{73}\right)3^n - \frac{4}{73}(8)^n - \frac{2n}{5}(3)^{n-1}$$



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Answer Key

Q1 (A)
Q2 (D)
Q3 (B)
Q4 (D)
Q5 (B, C)

Q6 (A)
Q7 (C)
Q8 (A)
Q9 (C)
Q10 (A)

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Hints & Solutions

Q1 Text Solution:

$$B_n = B_{n-1} + B_{n-2} + 2^{n-2}, n \geq 2$$

$$B_0 = 0$$

$$B_1 = 0$$

$$B_2 = 1$$

Q2 Text Solution:

$$b_1 - b_0 = 20$$

$$b_2 - b_1 = 40$$

$$b_3 - b_2 = 80$$

$$\vdots$$

$$b_n - b_{n-1} = 2(b_{n-1} - b_{n-2})$$

$$b_n = 3b_{n-1} - 2b_{n-2}, n \geq 2$$

Q3 Text Solution:

$$a_n = 2a_{n-1} + 3$$

$$= 2(2a_{n-2} + 3) + 3$$

$$= 2(2(2a_{n-3} + 3) + 3) + 3$$

$$\vdots$$

$$= 2^k a_{n-k} + 2^{k-1} \cdot 3 + 2^{k-2} \cdot 3 + \dots + 3$$

$$\text{Let } n - k = 0$$

$$\Rightarrow n = k$$

$$2^n \times 2 + 3(2^n - 1)$$

$$= 2^n(2 + 3) - 3 = 5 \cdot 2^n - 3$$

Q4 Text Solution:

$$a_n = a_{n-1} + 8a_{n-2} - 12a_{n-3}$$

Characteristic equation is,

$$t^3 - t^2 - 8t + 12 = 0$$

$$(t - 2)^2(t + 3) = 0$$

Complementary function,

$$a_n = (c_1 + c_2 n)2^n + c_3(-3)^n$$

$$a_0 = c_1 + c_3$$

$$0 = c_1 + c_3 \Rightarrow c_3 = -c_1$$

$$a_1 = (c_1 + c_2) \cdot 2 + c_3 \cdot (-3)$$

$$\Rightarrow 1 = 2c_1 + 2c_2 - 3c_3$$

$$\Rightarrow 1 = 5c_1 + 2c_2 \quad \dots(i)$$

$$a_2 = (c_1 + 2c_2) \cdot 4 + c_3 \cdot 9$$

$$\Rightarrow 3 = 4c_1 + 8c_2 + 9c_3$$

$$\Rightarrow 3 = -5c_1 + 8c_2 \quad \dots(ii)$$

using (i) and (ii)

$$c_1 = \frac{1}{25}, c_2 = \frac{2}{5}, c_3 = -\frac{1}{25}$$

$$a_n = \left(\frac{1}{25} + \frac{2n}{5}\right)2^n - \left(\frac{1}{25}\right)(-3)^n$$

Q5 Text Solution:

$$a_n = 11a_{n-1} - 30a_{n-2}$$

$$t^2 - 11t + 30 = 0$$

$$(t - 6)(t - 5) = 0$$

$$t = 6, 5$$

CF is

$$a_n = C_1(6)^n + C_2(5)^n$$

$$a_0 = C_1 + C_2$$

$$1 = C_1 + C_2 \quad \dots(i)$$

$$a_1 = 6C_1 + 5C_2$$

$$5 = 6C_1 + 5C_2 \quad \dots(ii)$$

$$5 = 6C_1 + 5(1 - C_1)$$

$$5 = C_1 + 5$$

$$C_1 = 0$$

$$\text{Then, } C_2 = 1$$

$$a_n = 5^n$$

Q6 Text Solution:

$$\text{Put } n = n+2$$

$$a_{n+1} + 7a_{n+1} - 18a_n = 2^{n+2} + 1$$

$$\phi(E) = E^2 + 7E - 18$$

$$\Rightarrow \phi(t) = t^2 + 7t - 18$$

$$\Rightarrow t = 2, -9$$

$$a_n^{(H)} = C_1(2)^n + C_2(-9)^n$$

$$\text{Now, } f(n) = 2^{n+1}$$

$$F(n) = 2^{n+2} + 1 = 4(2^n) + 1$$

$$\text{PS} = \text{Sol}^n \text{ of } \left(\frac{2^{n+2} + 1}{(t-2)(t+9)} \right)$$

$$= 4 \left(\frac{2^n}{(t-2)(t+9)} \right) + \frac{1^n}{(t-2)(t+9)}$$

$$= 4 \times \frac{1}{2+9} \times ({}^nC_1 2^{n-1}) + \frac{1^n}{(-1) \cdot (10)}$$

$$a_n = \frac{4n}{11} 2^{n-1} - \frac{1}{10}$$

Q7 Text Solution:

$$a_n - 6a_{n-1} + 9a_{n-2} = 3^n$$

$$\phi(E) = E^2 - 6E + 9$$

$$\phi(t) = t^2 - 6t + 9$$

$$\Rightarrow t = 3, 3$$

$$a_n^{(H)} = (C_1 + C_2 n)3^n$$

$$F(n) = 3^{n+2} = 9 \cdot 3^n$$

$$a_n^{(P)} = 9 \cdot \text{Sol of } \left\{ \frac{3^n}{\phi(E)} \right\} [b = 3]$$



$$\phi(b) = \phi(3) = 0$$

$$\phi(E) = (E - b)^m \cdot \psi(E)$$

$$\Rightarrow (E - 3)^2 = (E - 3)^2 \cdot \psi(E)$$

$$\Rightarrow \psi(E) = 1$$

$$PS = {}^nC_2(3)^{n-2}$$

$$a_n = a_n^{(H)} + a_n^{(P)}$$

$$a_n = (C_1 + C_2 n)3^n + {}^nC_2 3^{n-2}$$

$$C_1 = 1, C_2 = \frac{-1}{3}$$

$$a_n = \left(1 - \frac{n}{3}\right)3^n + {}^nC_2 3^{n-2}$$

$$a_n = (3 - n)3^{n-1} + {}^nC_2 3^{n-2}$$

$$a_n = 3^{n-2} [9 - 3n + {}^nC_2]$$

Q8 Text Solution:

$$\phi(E) = E^2 - 2E + 1$$

$$\Rightarrow \phi(t) = t^2 - 2t + 1$$

$$\Rightarrow t = 1, 1$$

$$a_n^{(H)} = C_1 + C_2 n$$

$$f(n) = n = 1^n (n)$$

$$\phi(1) = 0$$

$$a_n^{(P)} = 1^n (An + B)n^2 = (An + B)n^2$$

$$(An + B)n^2$$

$$- 2 \left\{ (A(n-1) + B)(n-1)^2 \right\}$$

$$+ \left[\{A(n-2) + B\}(n-2)^2 \right] = n$$

$$\text{Put } n=0, -3A + B = 0$$

$$\text{Put } n=1, 2B = 1$$

$$A = \frac{1}{6}, B = \frac{1}{2}$$

$$a_n^{(P)} = \left(\frac{n}{6} + \frac{1}{2}\right)n^2$$

$$a_n = C_1 + C_2 n + \left(\frac{n}{2} + \frac{1}{6}\right)n^2$$

$$\text{For } a_0, C_1 = 1$$

$$\text{For } a_1, C_2 = 1/3$$

$$a_n = 1 + \frac{n}{3} + \left(\frac{n}{6} + \frac{1}{2}\right)n^2$$

Q9 Text Solution:

$$\phi(E) = E^2 - 3E - 10$$

$$\phi(t) = t^2 - 3t - 10$$

$$\Rightarrow t = 5, -2$$

$$a_n^{(H)} = C_1(5)^n + C_2(-2)^n$$

$$f(n) = 5^n (2n + 1)$$

$$\phi(b) = 0$$

$$a_n^{(P)} = 5^n (An + B)n$$

$$\{5^n (An + B)n\} - [10 \cdot 5^{n-1}]$$

$$+ [3 \cdot 5^{n-1} \{A(n-1) + B\} \cdot (n-1)]$$

$$\text{Put } n=0, -5A + B = 0$$

$$\text{Put } n=1, 3A + 7B = 1$$

$$A = \frac{1}{38}, B = \frac{5}{38}$$

$$a_n = C_1(5)^n + C_2(-2)^n + 5^n (An + B)n$$

$$a_n = \left(\frac{156}{133}\right)5^n - \frac{23}{133}(-2)^n + 5^n \left(\frac{n}{38} + \frac{5}{38}\right)n$$

Q10 Text Solution:

$$\phi(E) = E^2 - 11E + 24$$

$$\phi(t) = t^2 - 11t + 24$$

$$\Rightarrow t = 3, 8$$

$$a_n^{(H)} = C_1(3)^n + C_2(8)^n$$

$$F(n) = 3^{n+2} = 9 \cdot 3^n$$

$$a_n^{(P)} = 9 \cdot \text{Sol of } \left\{ \frac{3^n}{\phi(E)} \right\}$$

$$\phi(3) = 0$$

$$\phi(E) = (E - 3)\psi(E)$$

$$\Rightarrow (E - 3)(E - 8) = (E - 3)\psi(E)$$

$$\Rightarrow \psi(E) = E - 8$$

$$\psi(3) = -5$$

$$a_n^{(P)} = \frac{-1}{5} \{ {}^nC_1 3^{n-1} \}$$

$$a_n = C_1(3)^n + C_2(8)^n - \frac{n}{5} 3^{n-1}$$

$$\text{For } a_0, 1 = C_1 + C_2$$

$$\text{For } a_1, 2 = 3C_1 + 8C_2 - \frac{1}{5}$$

$$C_1 = \frac{77}{73}, C_2 = \frac{-4}{73}$$

$$a_n = \left(\frac{77}{73}\right)3^n - \frac{4}{73}(8)^n - \frac{n}{5}(3)^{n-1}$$


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